

Matrix algebra

A *vector* is a list of numbers: $\mathbf{a} = (2, 4, 5, 1)$

$$\mathbf{a} = \begin{bmatrix} 2 \\ 4 \\ 5 \\ 1 \end{bmatrix}$$

The i th component of $\mathbf{a}$ is denoted a_i (above, $a_3 = 5$)

A vector with n components is an n –vector

The set of all n –vectors with real numbers as components is $\mathbb{R}^n$

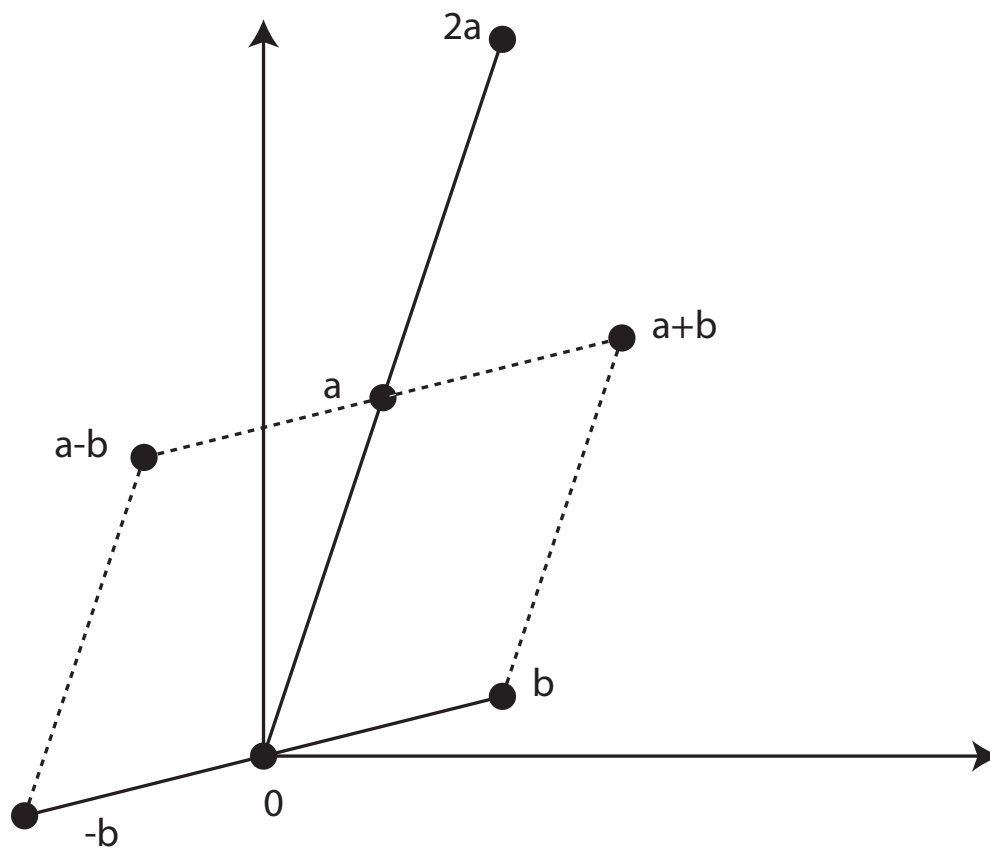
Vector arithmetic is very similar to ordinary arithmetic

Certain operations impose a restriction on the number of components the vectors can have (e.g. $\mathbf{a} + \mathbf{b}$ requires $\mathbf{a}$ and $\mathbf{b}$ to be the same length)

$$\text{Addition: } \begin{bmatrix} a_1 \\ a_2 \\ \dots \\ a_n \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \\ \dots \\ b_n \end{bmatrix} = \begin{bmatrix} a_1 + b_1 \\ a_2 + b_2 \\ \dots \\ a_n + b_n \end{bmatrix}$$

$$\text{Scalar multiplication: } \lambda \begin{bmatrix} a_1 \\ a_2 \\ \dots \\ a_n \end{bmatrix} = \begin{bmatrix} \lambda a_1 \\ \lambda a_2 \\ \dots \\ \lambda a_n \end{bmatrix}$$

Vector arithmetic geometrically



Linear dependence

A linear combination of vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}, \dots$ is a new vector of the form

$$\alpha \mathbf{a} + \beta \mathbf{b} + \gamma \mathbf{c} + \dots$$

where α, β, γ are scalars (elements of $\mathbb{R}$) and can potentially equal zero

A set of k vectors $\mathbf{a}^1, \mathbf{a}^2, \dots, \mathbf{a}^k$ is *linearly dependent* if it is possible to express one of the vectors as a linear combination of the others

The set is *linearly independent* if it's *not* possible to express one as a linear combination of the others

Example:

$$\mathbf{a} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \mathbf{c} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

These vectors are linearly dependent because $\mathbf{b} = \frac{1}{5}(\mathbf{a} + \mathbf{c})$

Criterion for linear dependence: $\mathbf{a}^1, \mathbf{a}^2, \dots, \mathbf{a}^k$ are *linearly dependent* if there exist scalars $\alpha_1, \alpha_2, \dots, \alpha_k$ that are *not all equal to zero* such that

$$\alpha_1 \mathbf{a}^1 + \alpha_2 \mathbf{a}^2 + \dots + \alpha_n \mathbf{a}^n = 0$$

(In the above example, $\alpha_a = \alpha_c = \frac{1}{5}$ and $\alpha_b = -1$)

How to test for linear dependence

We can solve for the scalars explicitly:

$$\mathbf{a} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \mathbf{c} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

$$2\alpha_a + \alpha_b + 3\alpha_c = 0$$

$$3\alpha_a + \alpha_b + 2\alpha_c = 0$$

$$-2\alpha_a - 3\alpha_c = \alpha_b$$

$$3\alpha_a - 2\alpha_a - 3\alpha_c + 2\alpha_c = 0$$

$$\alpha_a = \alpha_c \text{ and } \alpha_b = -5\alpha_a$$

(The vectors are linearly dependent)

Example of linear independence

$$\mathbf{a} = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \mathbf{c} = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

$$\alpha_a + \alpha_b + 3\alpha_c = 0$$

$$3\alpha_a + \alpha_b + 2\alpha_c = 0$$

$$\alpha_a + \alpha_b + \alpha_c = 0$$

$$-\alpha_a - 3\alpha_c = \alpha_b$$

$$3\alpha_a - \alpha_a - 3\alpha_c + 2\alpha_c = 0$$

$$2\alpha_a = \alpha_c$$

$$\alpha_a - 7\alpha_a + 2\alpha_a = 0$$

$$\alpha_a = \alpha_b = \alpha_c = 0$$

Matrices

A matrix is a rectangular array of numbers; a matrix with m rows and n columns is called an $m \times n$ matrix

The entry in the i th row and j th column of a matrix is the (i, j) entry of the matrix

$$\mathbf{A} = \begin{bmatrix} 2 & 3 & 5 & 2 & 1 \\ 3 & 16 & 1 & 4 & 5 \\ 0 & 6 & 1 & 10 & 8 \end{bmatrix}$$

$\mathbf{A}$ is a 3×5 matrix, with $a_{25} = 5$, $a_{34} = 10$, etc.

Matrix arithmetic

Addition and scalar multiplication work the same as for vectors:

$$\begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 3 \end{bmatrix} + \begin{bmatrix} 10 & 12 & 2 \\ 10 & 0 & 30 \end{bmatrix} = \begin{bmatrix} 11 & 14 & 5 \\ 11 & 3 & 33 \end{bmatrix}$$

$$5 \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 3 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 5 & 15 & 15 \end{bmatrix}$$

Matrix multiplication

Example: A 2×3 matrix, $\mathbf{A}$, and a 3-vector $\mathbf{x} = (x_1, x_2, x_3)$

$$\mathbf{A} = \begin{bmatrix} 6 & 8 & 7 \\ 5 & 1 & 9 \end{bmatrix} \text{ and } \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$\mathbf{Ax}$ is the following 2-vector:

$$\begin{bmatrix} 6 & 8 & 7 \\ 5 & 1 & 9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 6x_1 + 8x_2 + 7x_3 \\ 5x_1 + x_2 + 9x_3 \end{bmatrix}$$

$$\begin{bmatrix} 6 & 8 & 7 \\ 5 & 1 & 9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 6x_1 + 8x_2 + 7x_3 \\ 5x_1 + x_2 + 9x_3 \end{bmatrix}$$

Points to note

- The system of equations

$$6x_1 + 8x_2 + 7x_3 = b_1$$

$$5x_1 + x_2 + 9x_3 = b_2$$

can be written as $\mathbf{Ax} = \mathbf{b}$, with $\mathbf{b} = (b_1, b_2)$

- The 2×3 matrix $\mathbf{A}$ turns a 3-vector into a 2-vector; we can think of $\mathbf{A}$ as a mapping from $\mathbb{R}^3$ into $\mathbb{R}^2$

Matrix multiplication more generally

Let $\mathbf{A}$ be an $m \times n$ matrix and $\mathbf{B}$ be an $n \times s$ matrix; then $\mathbf{AB}$ is an $m \times s$ matrix with entry (i, j) equal to

$$(i, j) = \sum_{k=1}^n a_{ik} b_{kj}$$

Each entry (i, j) takes the sum of the product of each i th row entry in $\mathbf{A}$ with the j th column entry in $\mathbf{B}$

Example:

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \mathbf{B} = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}, \mathbf{AB} = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{bmatrix}$$

Bigger example:

$$\mathbf{A} = \begin{bmatrix} 1 & 5 \\ 2 & -3 \\ 4 & -8 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 2 & -7 & 3 & 0 \\ 1 & 0 & 4 & -5 \end{bmatrix}$$

$$\mathbf{AB} = \begin{bmatrix} 1 \times 2 + 5 \times 1 & 1 \times -7 + 5 \times 0 & 1 \times 3 + 5 \times 4 & 1 \times 0 + 5 \times -5 \\ 2 \times 2 - 3 \times 1 & 2 \times -7 - 3 \times 0 & 2 \times 3 - 3 \times 4 & 2 \times 0 - 3 \times -5 \\ 4 \times 2 - 8 \times 1 & 4 \times -7 - 8 \times 0 & 4 \times 3 - 8 \times 4 & 4 \times 0 - 8 \times -5 \end{bmatrix}$$

$$\mathbf{AB} = \begin{bmatrix} 7 & -7 & 23 & -25 \\ 1 & -14 & -6 & 15 \\ 0 & -28 & -20 & 40 \end{bmatrix}$$

Important facts about matrix multiplication

- We can multiply any two matrices $\mathbf{A}\mathbf{B}$ when the number of columns in $\mathbf{A}$ equals the number of rows in $\mathbf{B}$

$$\underbrace{\mathbf{A}}_{n \times m} \underbrace{\mathbf{B}}_{m \times k} = \underbrace{\mathbf{C}}_{n \times k}$$

- Matrix multiplication is NOT commutative

$$\mathbf{AB} \neq \mathbf{BA}$$

- First, both sides are only defined if $\mathbf{A}$ is $n \times m$ and $\mathbf{B}$ is $m \times n$
- Even if $\mathbf{A}$ is $m \times m$ and $\mathbf{B}$ is $m \times m$ they may still not be the same

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix}$$

The $n \times n$ identity matrix, $\mathbf{I}_n$

This is an $n \times n$ matrix whose (i, j) entry is 1 if $i = j$ and 0 otherwise

$$\mathbf{I}_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

If A is $n \times m$, then

$$\mathbf{I}_n \mathbf{A} = \mathbf{A}$$

and

$$\mathbf{A} \mathbf{I}_m = \mathbf{A}$$

Echelon matrices

An echelon matrix has the following two properties:

- 1 There's at least one non-zero entry and rows with all zeros lie below rows that have at least one non-zero entry
- 2 In each row after the first, the left-most non-zero entry lies to the right of the left-most non-zero entry in the preceding row

$$\begin{bmatrix} \star & \cdot & \cdot & \cdot \\ 0 & \star & \cdot & \cdot \\ 0 & 0 & 0 & \star \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

- The first non-zero entry in each row ($\star$) is the *pivot*

Gaussian elimination (creating echelon matrices to solve $\mathbf{Ax} = \mathbf{B}$)

The following are the elementary operations we'll use:

- Exchanging two rows
- Subtracting a multiple of one row from another

We will subtract multiples of one row (the pivot row) from each of the rows below it

We'll work with an augmented matrix

$$\left[\mathbf{A} \mid \mathbf{b} \right]$$

Suppose we want to solve $\mathbf{Ax} = \mathbf{B}$ for $\mathbf{x}$, with

$$\mathbf{A} = \begin{bmatrix} 2 & -4 & 1 & -8 \\ 4 & -8 & 7 & -6 \\ -1 & 2 & 1 & 7 \end{bmatrix} \text{ and } \mathbf{b} = \begin{bmatrix} -4 \\ 12 \\ 8 \end{bmatrix}$$

We start with augmented matrix

$$\left[\mathbf{A} \quad \mathbf{b} \right] = \begin{bmatrix} 2 & -4 & 1 & -8 & -4 \\ 4 & -8 & 7 & -6 & 12 \\ -1 & 2 & 1 & 7 & 8 \end{bmatrix}$$

$$[\mathbf{A} \quad \mathbf{b}] = \begin{bmatrix} 2 & -4 & 1 & -8 & -4 \\ 4 & -8 & 7 & -6 & 12 \\ -1 & 2 & 1 & 7 & 8 \end{bmatrix}$$

Using the top row as a pivot we subtract multiples of pivot from each lower row (2 and $-\frac{1}{2}$, respectively):

$$\begin{bmatrix} 2 & -4 & 1 & -8 & -4 \\ 0 & 0 & 5 & 10 & 20 \\ 0 & 0 & \frac{3}{2} & 3 & 6 \end{bmatrix}$$

Using second row as a pivot we multiply it by $\frac{3}{10}$ and subtract from third:

$$\begin{bmatrix} 2 & -4 & 1 & -8 & -4 \\ 0 & 0 & 5 & 10 & 20 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

We're left with:

$$\begin{bmatrix} 2 & -4 & 1 & -8 & -4 \\ 0 & 0 & 5 & 10 & 20 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

We have 2 equations and 4 unknowns; assign arbitrary values to $x_2 = \lambda$ and $x_4 = \mu$ (the entries without pivots)

$$5x_3 = 20 - 10\mu$$

$$x_3 = 4 - 2\mu$$

and

$$2x_1 = 4\lambda - (4 - 2\mu) + 8\mu - 4$$

$$x_1 = -4 + 2\lambda + 5\mu$$

Summarizing, we have

$$x_1 = -4 + 2\lambda + 5\mu$$

$$x_2 = \lambda$$

$$x_3 = 4 - 2\mu$$

$$x_4 = \mu$$

Which your book writes as

$$\mathbf{x} = \begin{bmatrix} -4 \\ 0 \\ 4 \\ 0 \end{bmatrix} + \lambda \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + \mu \begin{bmatrix} 5 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

Solve the system $\mathbf{Ax} = \mathbf{b}$ where

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 4 \\ 2 & 1 & 1 \\ 4 & 8 & 7 \\ 4 & 3 & 9 \end{bmatrix} \text{ and } \mathbf{b} = \begin{bmatrix} 5 \\ 8 \\ 5 \\ 8 \end{bmatrix}$$

Matrix rank

The rank of a matrix $\mathbf{A}$ is the maximum number of linearly independent columns of $\mathbf{A}$; it is also the maximum number of linearly independent rows of $\mathbf{A}$

This is the dimension (k) of the vector space $\mathbb{R}^k$ *spanned* by its columns (or rows)

The *span* of a set of a finite set of vectors S is the set of all linear combinations of the elements of S ; if this equals $\mathbb{R}^k$ then $\text{Span}(S) = \mathbb{R}^k$

Rank and span

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

Columns 1 and 2 are linearly independent, but their sum equals Column 3;
the rank of $\mathbf{A} = 2$

These three vectors span $\mathbb{R}^2$ but not $\mathbb{R}^3$: Show that for any $(x_1, x_2) \in \mathbb{R}^2$,
we can find a $\lambda_1, \lambda_2, \lambda_3$ such that

$$\lambda_1(1, 0, 1) + \lambda_2(0, 1, 1) + \lambda_3(1, 1, 2) = (x_1, x_2, x_1 + x_2)$$

They span $\mathbb{R}^2$ and the space of all vectors in $\mathbb{R}^3$ of form $(x_1, x_2, x_1 + x_2)$

Show that for any $(x_1, x_2, x_3) \in \mathbb{R}^3$, we *can't* find a $\lambda_1, \lambda_2, \lambda_3$ such that

$$\lambda_1(1, 0, 1) + \lambda_2(0, 1, 1) + \lambda_3(1, 1, 2) = (x_1, x_2, x_3)$$

Finding the rank of a matrix $\mathbf{A}$

- Reduce $\mathbf{A}$ to an echelon matrix $\mathbf{E}$ using Gaussian elimination
- The number of non-zero rows of $\mathbf{E}$ is the rank of $\mathbf{A}$!

Properties of ranks

- The rank of an $n \times m$ matrix $\mathbf{A}$ can't be greater than either n or m :
 $\text{Rank}(\mathbf{A}) \leq \min(n, m)$
- A matrix has *full rank* if its rank is the biggest it could be:
 $\text{Rank}(\mathbf{A}) = \min(n, m)$
- A square matrix is *invertible* if and only if it has full rank

Inverse Matrices

A square matrix $\mathbf{A}$ is *singular* if the system of equations $\mathbf{Ax} = \mathbf{0}$ has a non-zero solution for $\mathbf{x}$

$\mathbf{A}$ is *non-singular* if the system of equations $\mathbf{Ax} = \mathbf{0}$ has the unique solution $\mathbf{x} = \mathbf{0}$

$\mathbf{A}$ is non-singular if and only if it has full rank

Why we care: If $n \times n$ matrix $\mathbf{A}$ is non-singular and $\mathbf{y}$ is any n -vector, then there is exactly one n -vector $\mathbf{x}$ that solves

$$\mathbf{Ax} = \mathbf{y}$$

Square matrix $\mathbf{A}$ is invertible if there is a square matrix $\mathbf{A}^{-1}$ with the property that

$$\mathbf{Ax} = \mathbf{y} \text{ if and only if } \mathbf{x} = \mathbf{A}^{-1}\mathbf{y}$$

$\mathbf{A}^{-1}$ is the *inverse* of $\mathbf{A}$

Obviously if we know $\mathbf{y}$ then we can solve $\mathbf{Ax} = \mathbf{y}$ explicitly, just using Gaussian elimination

Inverses can be useful if we want to solve the above equation for *any* $\mathbf{y}$

How to invert a matrix

Take

$$\begin{bmatrix} \mathbf{A} & \mathbf{I} \end{bmatrix}$$

and use Gauss-Jordan elimination to convert it into

$$\begin{bmatrix} \mathbf{I} & \mathbf{A}^{-1} \end{bmatrix}$$

Gauss-Jordan elimination is similar to Gaussian elimination, but turns matrix into a diagonal matrix, instead of just an echelon matrix

Let's invert

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 2 \\ 3 & 1 & 1 \\ 3 & 1 & 2 \end{bmatrix}$$

$$\text{Step 1: } \begin{bmatrix} 2 & 1 & 2 & 1 & 0 & 0 \\ 3 & 1 & 1 & 0 & 1 & 0 \\ 3 & 1 & 2 & 0 & 0 & 1 \end{bmatrix}$$

$$\text{Step 2: } \begin{bmatrix} 2 & 1 & 2 & 1 & 0 & 0 \\ 0 & -\frac{1}{2} & -2 & -\frac{3}{2} & 1 & 0 \\ 0 & -\frac{1}{2} & -1 & -\frac{3}{2} & 0 & 1 \end{bmatrix}$$

$$\text{Step 2: } \begin{bmatrix} 2 & 1 & 2 & 1 & 0 & 0 \\ 0 & -\frac{1}{2} & -2 & -\frac{3}{2} & 1 & 0 \\ 0 & -\frac{1}{2} & -1 & -\frac{3}{2} & 0 & 1 \end{bmatrix}$$

Since we are creating a diagonal we want to get rid of the “1” in (1, 2)

$$\text{Step 3: } \begin{bmatrix} 2 & 0 & -2 & -2 & 2 & 0 \\ 0 & -\frac{1}{2} & -2 & -\frac{3}{2} & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix}$$

This added $2 \times \text{Row 2}$ to Row 1 and subtracted Row 2 from Row 3

$$\text{Step 3: } \begin{bmatrix} 2 & 0 & -2 & -2 & 2 & 0 \\ 0 & -\frac{1}{2} & -2 & -\frac{3}{2} & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix}$$

We get a diagonal matrix by adding $2 \times \text{Row 3}$ to Rows 1 & 2

$$\text{Step 4: } \begin{bmatrix} 2 & 0 & 0 & -2 & 0 & 2 \\ 0 & -\frac{1}{2} & 0 & -\frac{3}{2} & -1 & 2 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix}$$

We finish by normalizing

$$\text{Step 5: } \begin{bmatrix} 1 & 0 & 0 & -1 & 0 & 1 \\ 0 & 1 & 0 & 3 & 2 & -4 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix} \text{ and } \mathbf{A}^{-1} = \begin{bmatrix} -1 & 0 & 1 \\ 3 & 2 & -4 \\ 0 & -1 & 1 \end{bmatrix}$$

Inverse facts

- $\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$
- $(\mathbf{A}^{-1})^{-1} = \mathbf{A}$
- If $\mathbf{A}$ and $\mathbf{B}$ are invertible so is $\mathbf{AB}$ and $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$
- If $\mathbf{A}$ is 2×2 with

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

and $ad \neq bc$ then

$$\mathbf{A}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

(If $ad = bc$ then $\mathbf{A}$ isn't invertible)